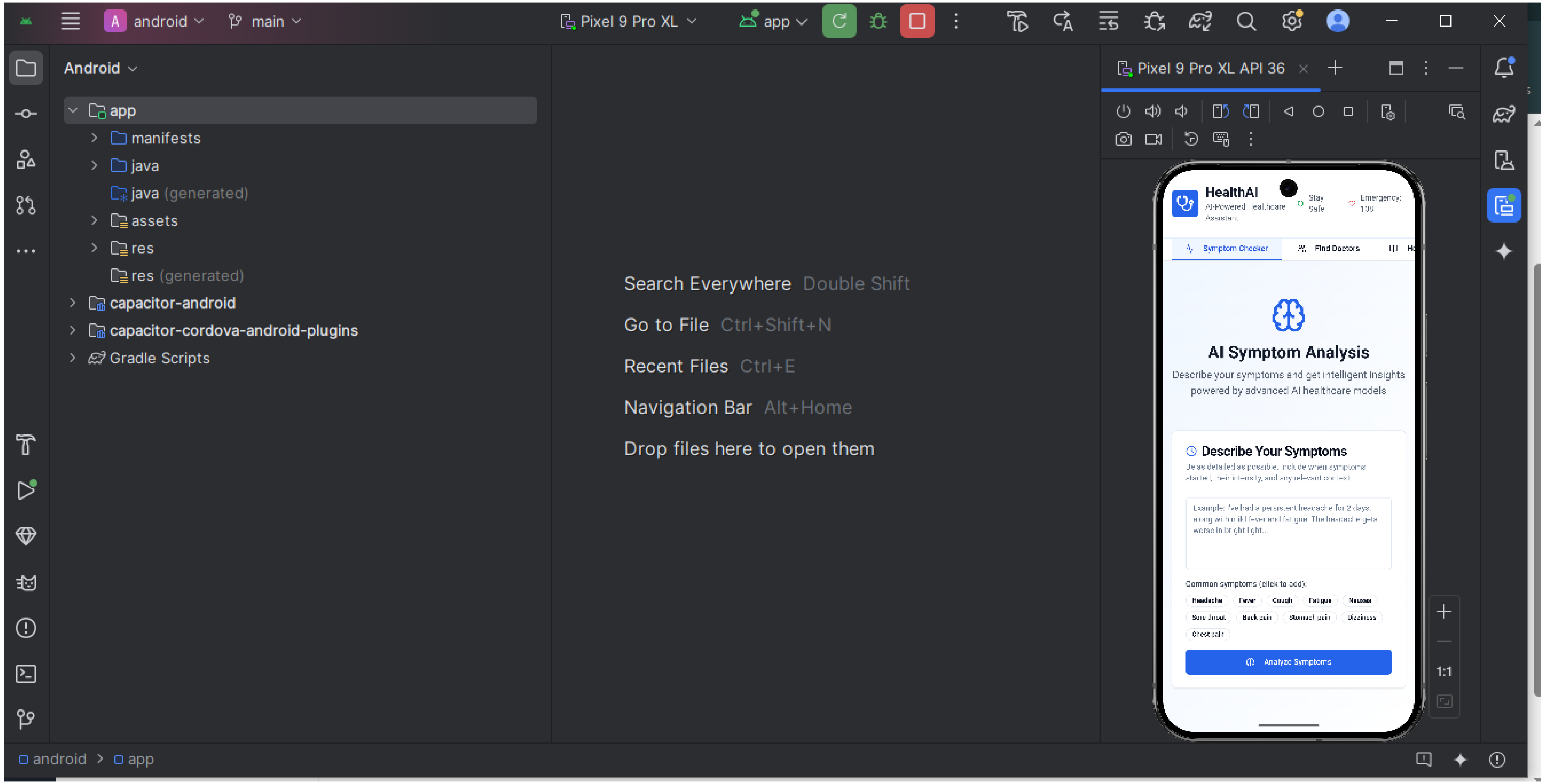
Team Name: Ctrl Alt Elite
Topic : Doctor AI



Abstract

- **Objective / Aim:**
  The objective of *DoctorAI* is to create an intelligent, AI-powered system that can accurately understand users' health queries, provide verified medical insights, and guide them toward appropriate next steps, such as consulting a doctor or scheduling a health checkup.

- **Proposed Solution / Approach:**
  DoctorAI leverages Natural Language Processing (NLP) and Large Language Model (LLM) technology to interpret user queries in natural language. The system integrates a curated medical knowledge base with machine learning models to generate reliable, explainable responses. It can process doctor-patient conversation datasets, classify symptoms, and match them with possible conditions using confidence-based reasoning. The platform can be deployed as a chatbot or integrated with telemedicine portals for real-time support. Future enhancements include multilingual capabilities and integration with wearable health data.

- **Expected Impact / Outcome:**
  DoctorAI will serve as a virtual pre-diagnosis assistant, improving early health awareness, reducing misinformation, and supporting overburdened healthcare systems. It empowers users with trustworthy information, facilitates quick triage, and helps medical professionals focus on critical cases. In the long run, the solution aims to bridge the healthcare accessibility gap through intelligent, inclusive, and data-driven support.

Problem Understanding

- **Limited Access to Medical Guidance:**
  Many people struggle to get quick and reliable basic medical guidance due to long hospital queues, lack of nearby doctors, or high consultation fees.
- **Information Overload & Inaccuracy:**
  Users often Google symptoms and receive confusing, unreliable, or overly generic information, which may cause panic or wrong assumptions.
- **Delayed First-Level Diagnosis:**
  Minor health concerns escalate because people don't know whether the situation needs immediate care, home remedies, or a specialist visit.
- **Lack of Personalized Suggestions:**
  Existing symptom-checker apps give generic outputs and fail to consider user's age, condition, history, or context.

Objective of the Project

- **Provide instant, reliable first-level medical guidance** using AI to help users understand symptoms and possible causes.
- **Simplify medical reports** (CBC, LFT, KFT, lipid profile, prescriptions, etc.) by explaining results in easy-to-understand language.
- **Reduce unnecessary hospital visits** by telling users whether a condition is mild, moderate, or needs immediate medical attention.
- **Assist doctors and clinics** by filtering common queries and reducing repetitive questions.
- **Offer personalized healthcare advice** based on age, symptoms, vitals, and user context.
- **Bridge the healthcare access gap** for rural, remote, or low-resource areas with AI-powered support.
- **Promote preventive healthcare** through reminders, risk alerts, and lifestyle suggestions.
- **Provide multilingual medical assistance** so users can understand health information in their preferred language.
- **Act as a smart companion** that stores and analyzes previous interactions, symptoms, and report history.

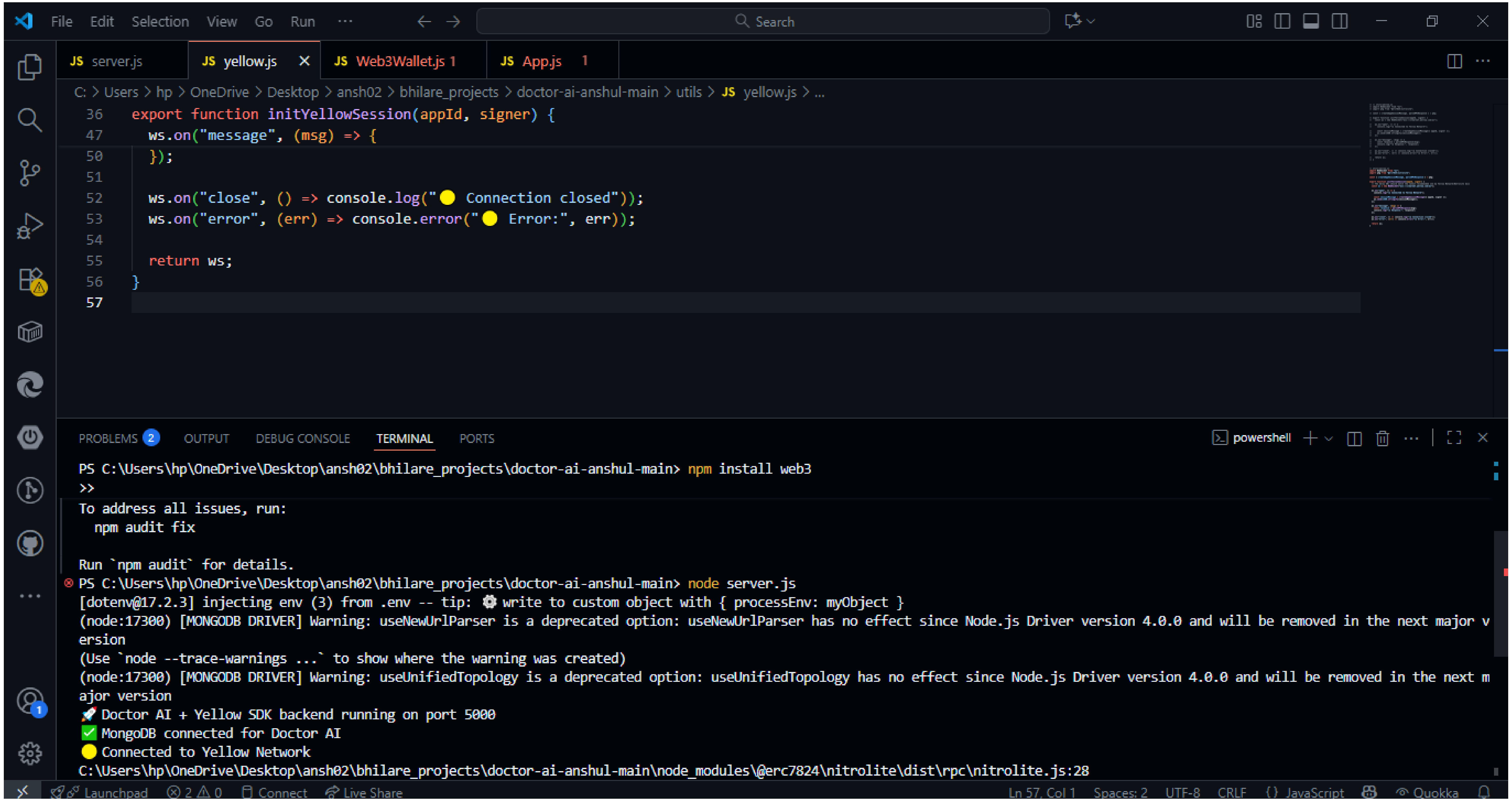
Solution Design & Innovation

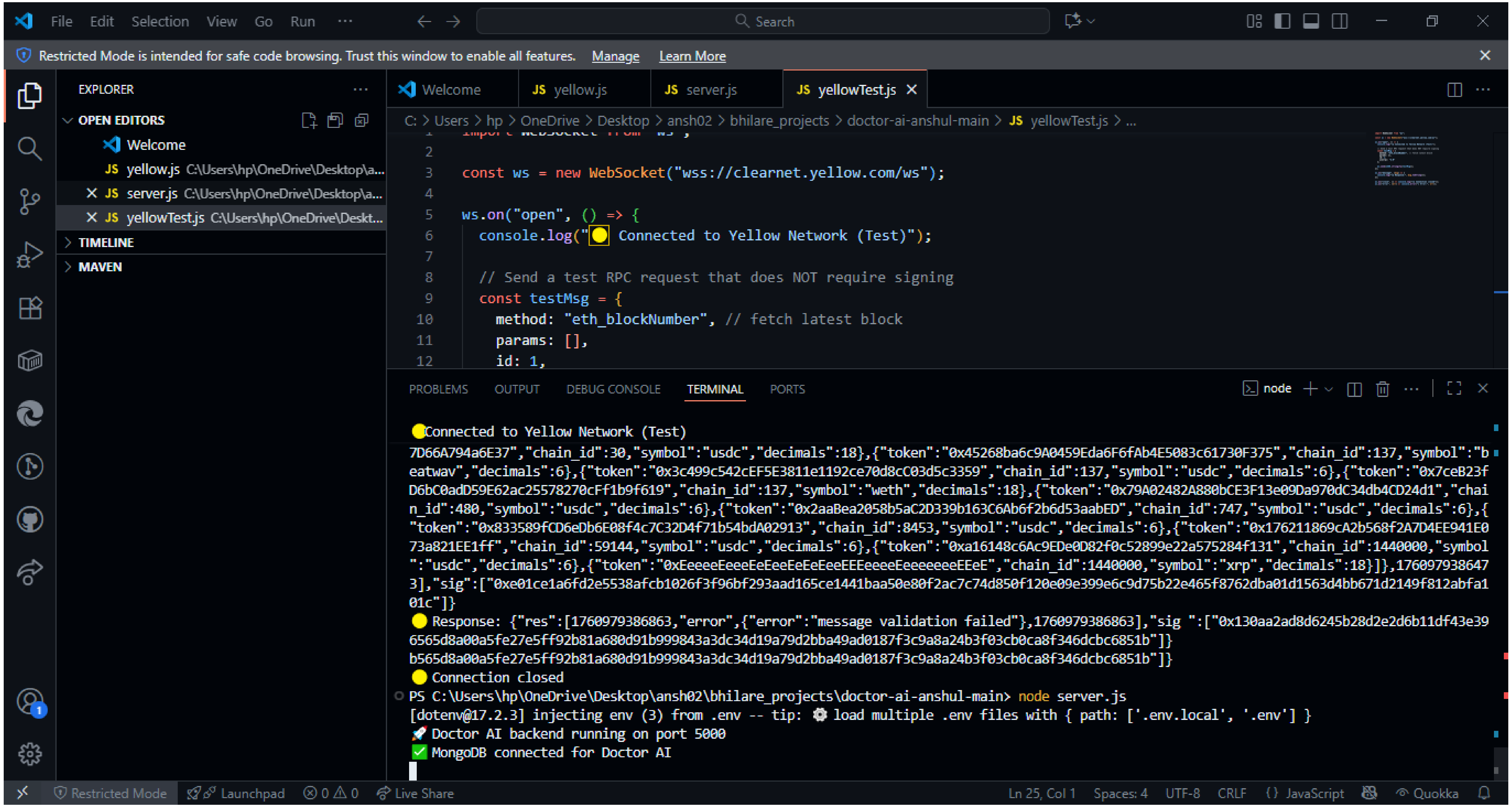
- **Frontend:** Chat UI with voice, report upload, history, multilingual support.
- **Backend:** Node.js API orchestrating AI, OCR & rule engine.
- **AI Layer:**
- LLM for conversation & explanations.
- OCR for report extraction.
- Rule-based triage for safety (red/yellow/green).
- **Database:** Encrypted user data, report storage, medical knowledge base.
- **Admin Panel:** Doctors view flagged cases & summaries.
- **Hybrid safety-first architecture** — LLM creativity + deterministic medical rules ensures better safety than pure chatbots.
- **Report auto-explain with context-aware ranges** — explains labs with age/sex-specific ranges and simple actions.
- **Conversational escalation with pre-filled doctor summary** — saves doctor time and improves triage efficiency.

Solution Workflow / Approach

- **User Entry**
- Inputs: free-text chat, structured symptom form, voice, or upload (image/PDF of reports).
- Quick pre-check: collect age, sex, major conditions, allergies, current meds.
- **Pre-processing**
- Sanitize text, normalize units, run language detection/translations.
- If voice → speech-to-text; if image/PDF → OCR.
- **Information Extraction**
- Parse OCR → identify test names, values, units, reference ranges.
- Extract symptom entities, durations, severity from text (NER + regex).
- **Context Assembly**
- Build a structured context bundle: user profile, vitals reported, extracted lab values, symptom timeline, recent history.
- **Deterministic Safety Check (Rule Engine)**
- Apply hard safety rules (red flags). Example: chest pain + breathlessness → urgent escalation.

Architecture Diagram

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const ws = new WebSocket("wss://clearnet.yellow.com/ws");
ws.on("open", () => {
console.log("🟡 Connected to Yellow Network (Test)");
// Send a test RPC request that does NOT require signing
const testMsg = {
method: "eth_blockNumber", // fetch latest block
params: [],
id: 1,
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
node
🟡Connected to Yellow Network (Test)
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🟡 Connection closed
PS C:\Users\hp\OneDrive\Desktop\ansh02\bhilare_projects\doctor-ai-anshul-main> node server.js
[dotenv@17.2.3] injecting env (3) from .env -- tip: ⚙️ load multiple .env files with { path: ['.env.local', '.env'] }
🚀 Doctor AI backend running on port 5000
✅ MongoDB connected for Doctor AI
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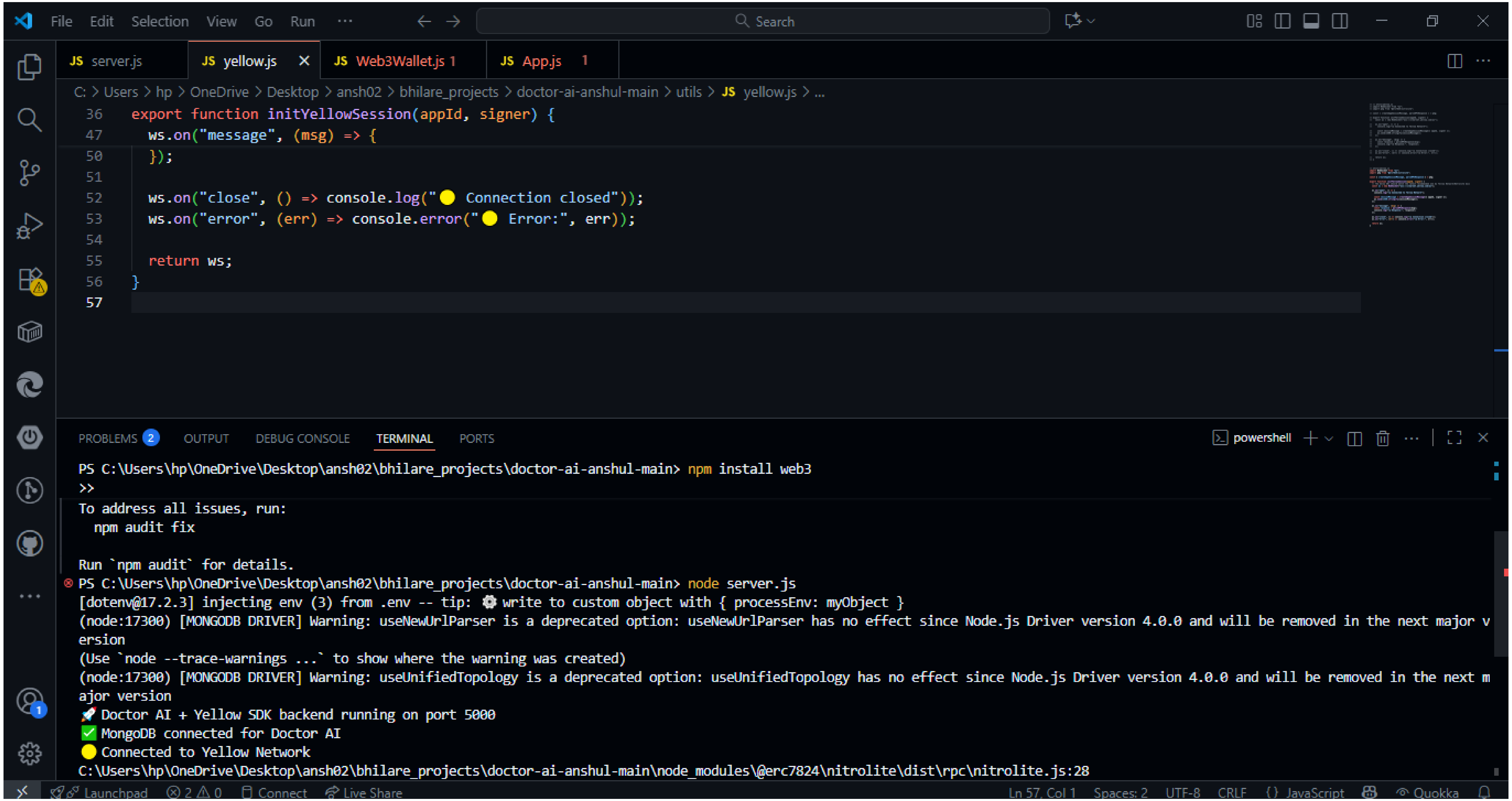
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export function initYellowSession(appId, signer) {
ws.on("message", (msg) => {
});
ws.on("close", () => console.log("🟡 Connection closed"));
ws.on("error", (err) => console.error("🟡 Error:", err));
return ws;
}
PROBLEMS 2
OUTPUT
DEBUG CONSOLE
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powershell
PS C:\Users\hp\OneDrive\Desktop\ansh02\bhilare_projects\doctor-ai-anshul-main> npm install web3
>>
To address all issues, run:
npm audit fix
Run `npm audit` for details.
PS C:\Users\hp\OneDrive\Desktop\ansh02\bhilare_projects\doctor-ai-anshul-main> node server.js
[dotenv@17.2.3] injecting env (3) from .env -- tip: ⚙️ write to custom object with { processEnv: myObject }
(node:17300) [MONGODB DRIVER] Warning: useNewUrlParser is a deprecated option: useNewUrlParser has no effect since Node.js Driver version 4.0.0 and will be removed in the next major v
ersion
(Use `node --trace-warnings ...` to show where the warning was created)
(node:17300) [MONGODB DRIVER] Warning: useUnifiedTopology is a deprecated option: useUnifiedTopology has no effect since Node.js Driver version 4.0.0 and will be removed in the next m
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✅ MongoDB connected for Doctor AI
🟡 Connected to Yellow Network
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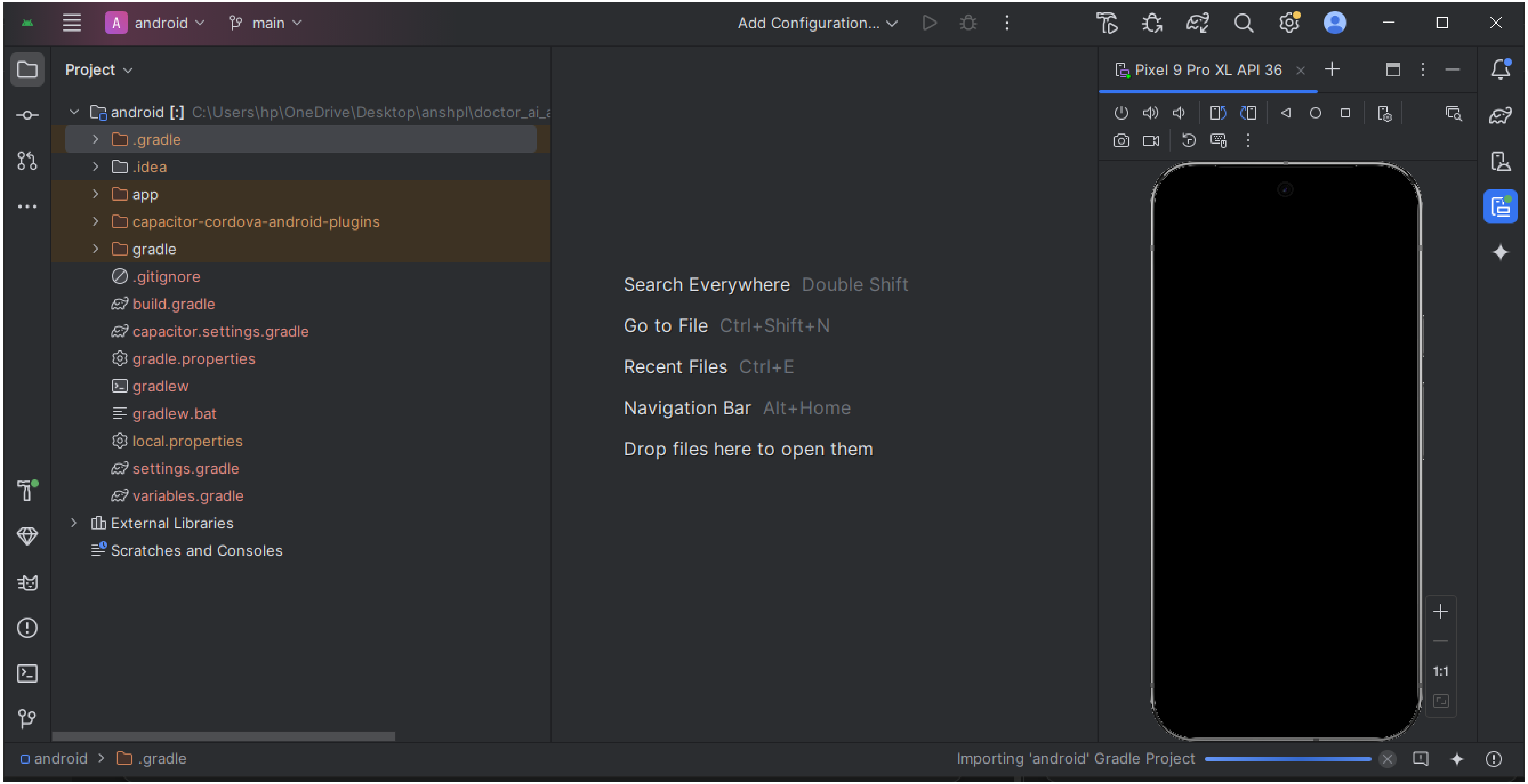
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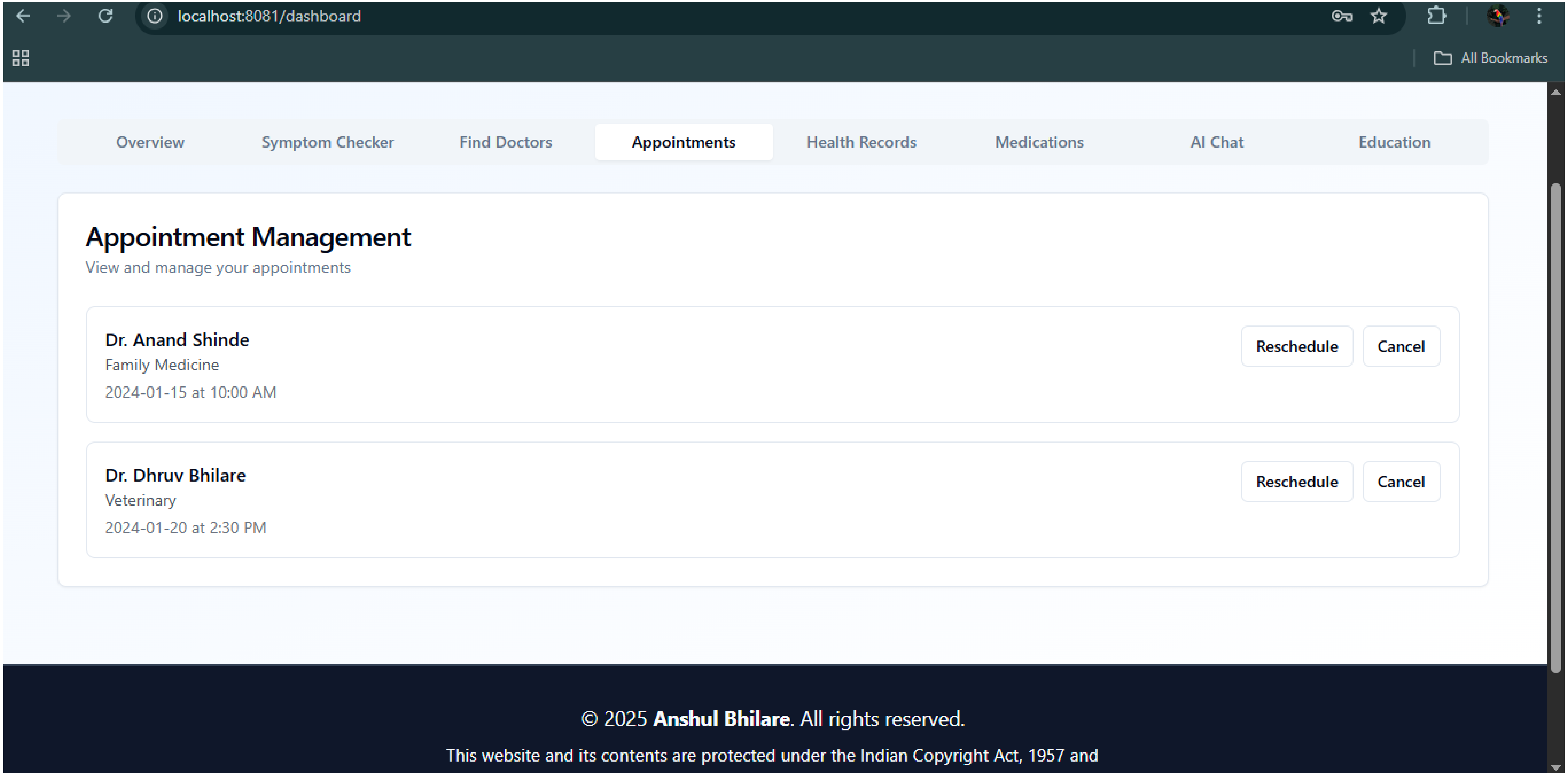
localhost:8081/dashboard
All Bookmarks
Overview
Symptom Checker
Find Doctors
Appointments
Health Records
Medications
AI Chat
Education
Appointment Management
View and manage your appointments
Dr. Anand Shinde
Family Medicine
2024-01-15 at 10:00 AM
Reschedule
Cancel
Dr. Dhruv Bhilare
Veterinary
2024-01-20 at 2:30 PM
Reschedule
Cancel
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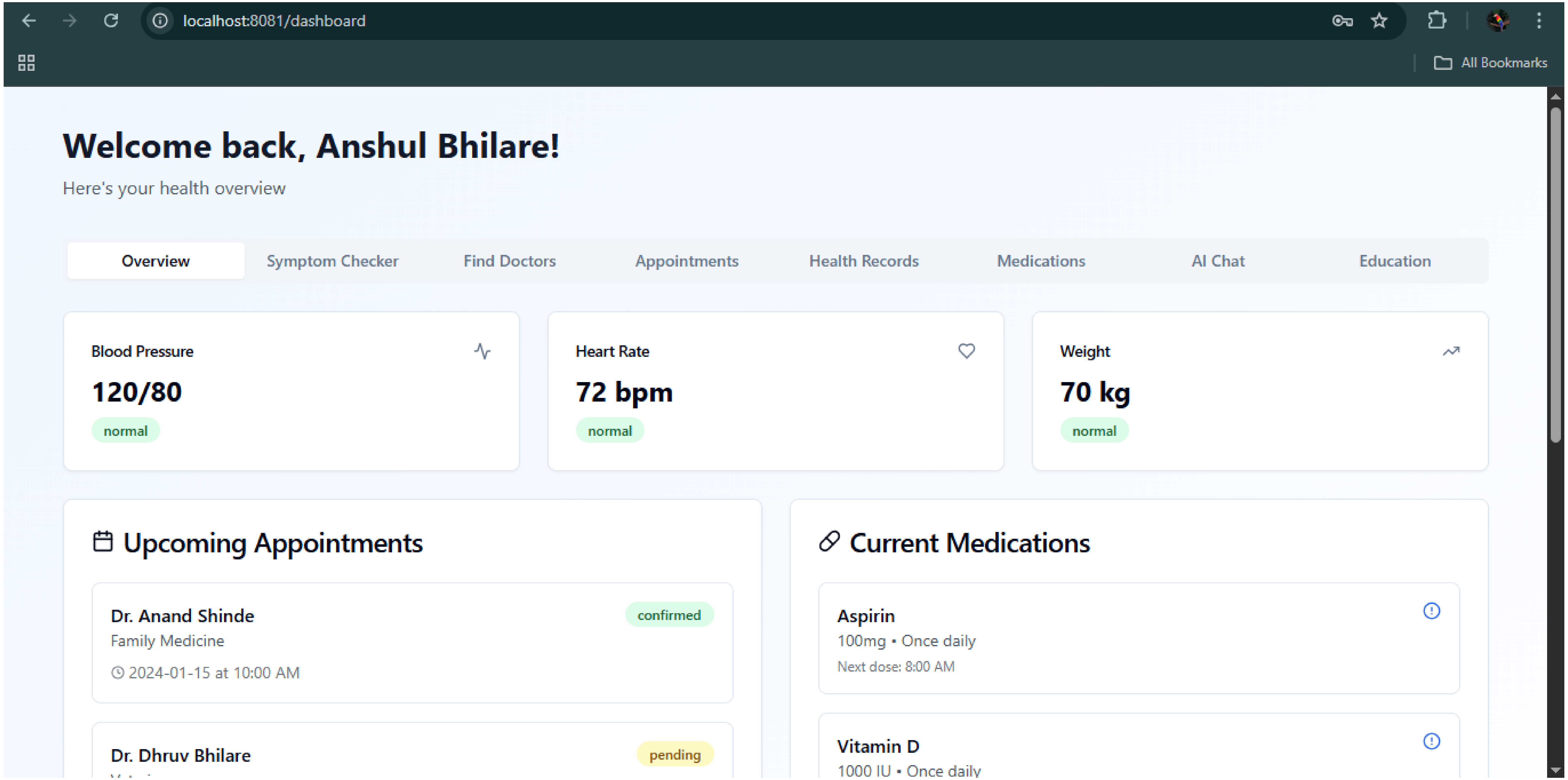
localhost:8081/dashboard
All Bookmarks
Welcome back, Anshul Bhilare!
Here's your health overview
Overview
Symptom Checker
Find Doctors
Appointments
Health Records
Medications
AI Chat
Education
Blood Pressure
120/80
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Heart Rate
72 bpm
normal
Weight
70 kg
normal
Upcoming Appointments
Dr. Anand Shinde
confirmed
Family Medicine
2024-01-15 at 10:00 AM
Dr. Dhruv Bhilare
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Current Medications
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100mg • Once daily
Next dose: 8:00 AM
Vitamin D
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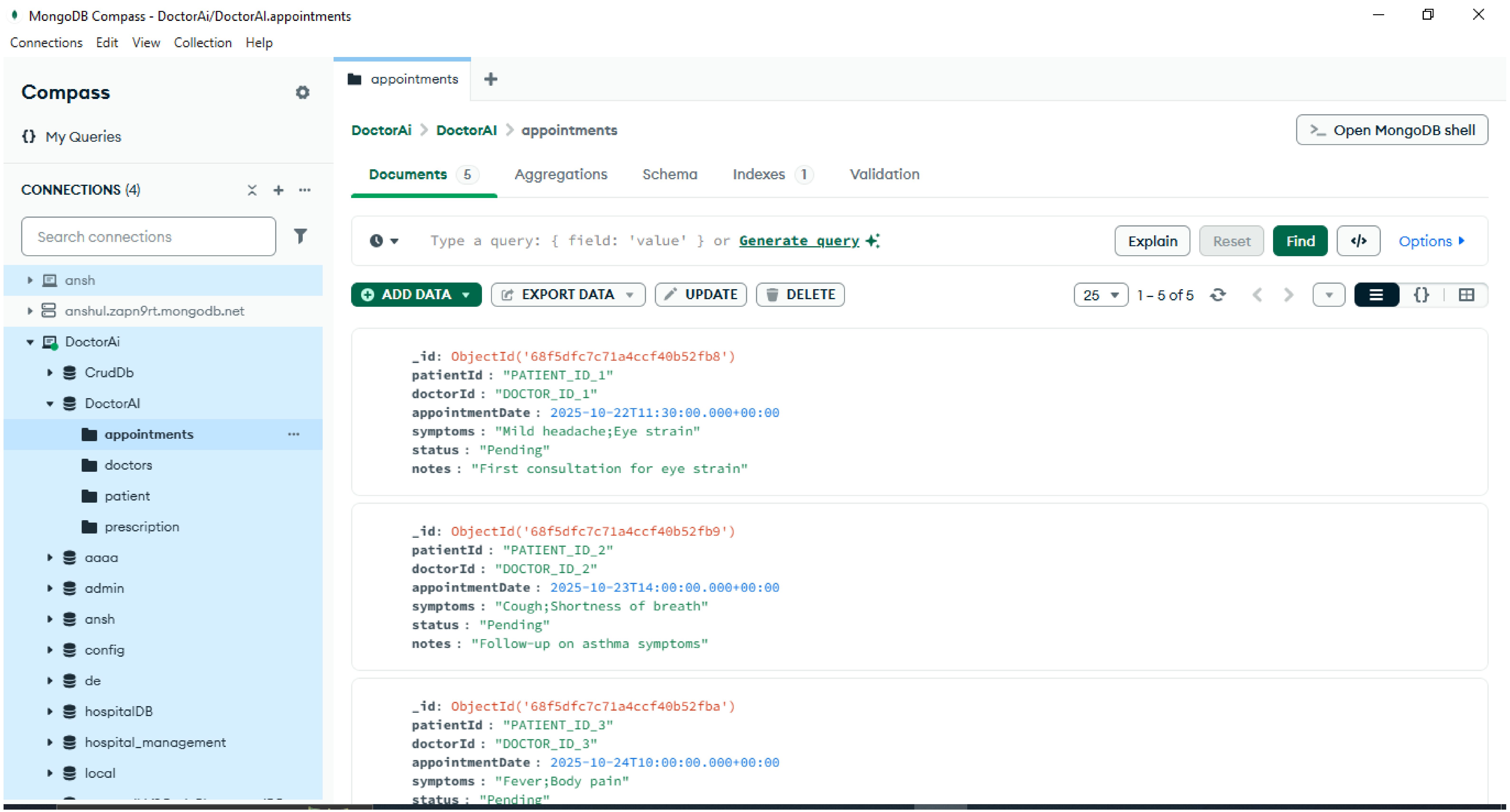
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25
1 – 5 of 5
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notes : "First consultation for eye strain"
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patientId : "PATIENT_ID_2"
doctorId : "DOCTOR_ID_2"
appointmentDate : 2025-10-23T14:00:00.000+00:00
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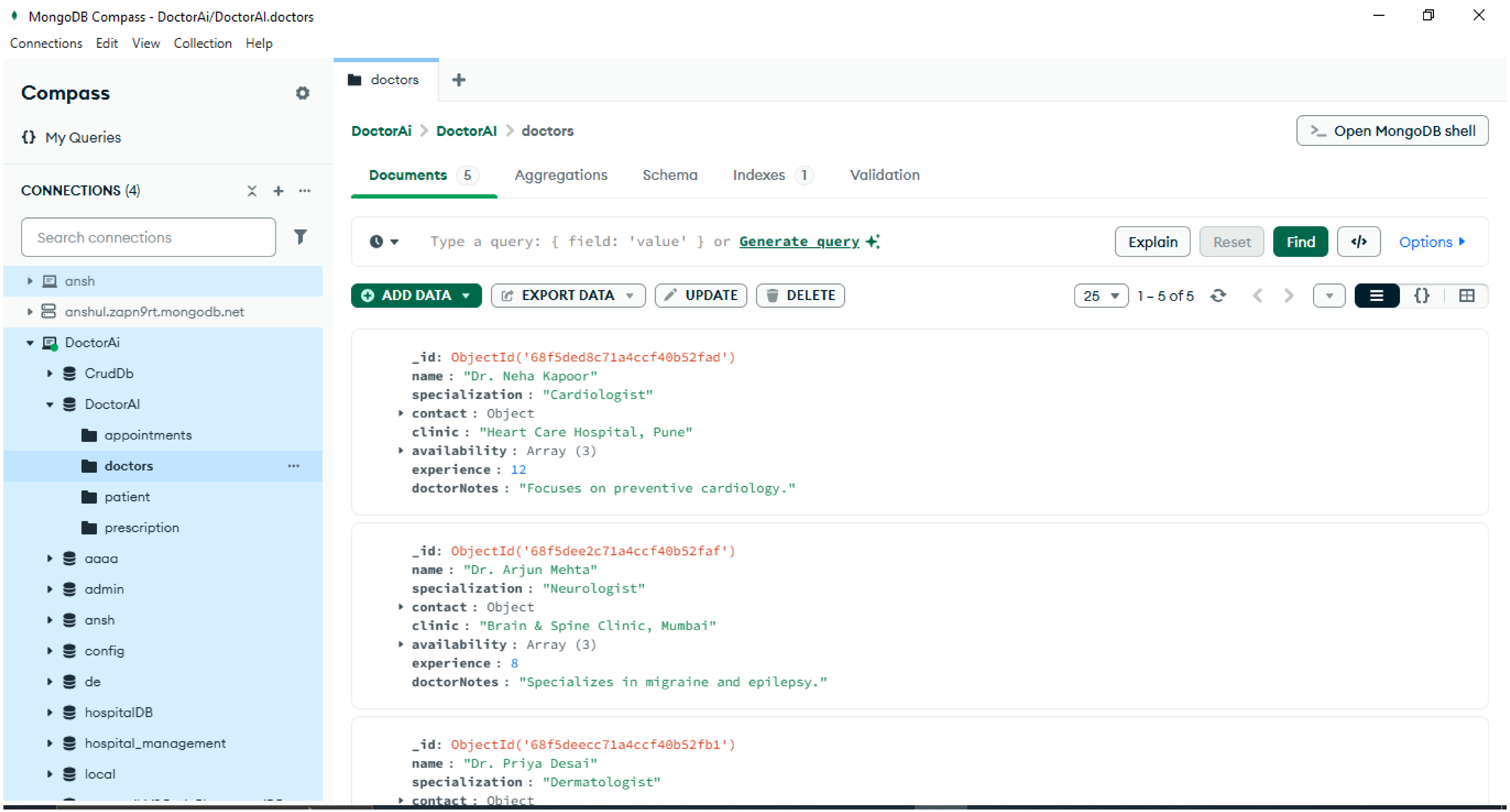
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Type a query: { field: 'value' } or Generate query
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25
1 – 5 of 5
_id: ObjectId('68f5ded8c71a4ccf40b52fad')
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experience : 12
doctorNotes : "Focuses on preventive cardiology."
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name : "Dr. Arjun Mehta"
specialization : "Neurologist"
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clinic : "Brain & Spine Clinic, Mumbai"
availability : Array (3)
experience : 8
doctorNotes : "Specializes in migraine and epilepsy."
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name : "Dr. Priya Desai"
specialization : "Dermatologist"
contact : Object

Key Features

- **Intelligent Symptom Analysis**
- Users enter symptoms in natural language; AI identifies possible causes and urgency levels.
- Quick triage: **Green (mild), Yellow (moderate), Red (critical)**.
- **2. Lab Report Auto-Explanation**
- Upload any medical report (CBC, LFT, KFT, Lipid, Thyroid, etc.).
- OCR extracts values → AI explains results in simple, easy language.
- **3. Prescription Understanding**
- AI reads doctor prescriptions and explains medicines, dosage, and precautions.
- Highlights interactions and contradictions based on user history.
- **4. AI-Generated Personalized Health Advice**
- Gives custom suggestions based on age, symptoms, history, and lifestyle.
- Helps users decide: **home care vs. test required vs. doctor visit**.
- **5. Multilingual Support**
- Assistant can answer in multiple Indian languages for better accessibility.

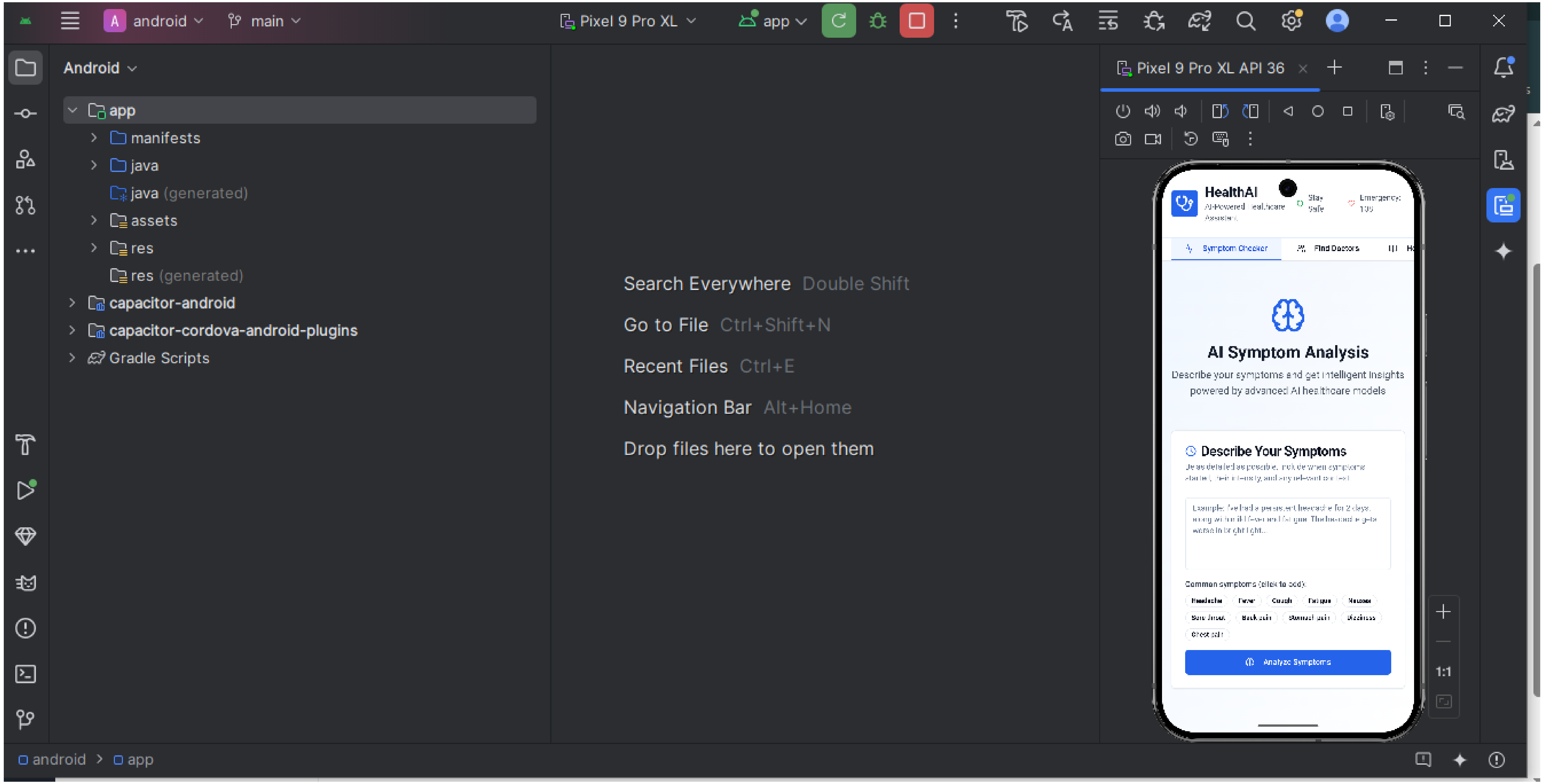
Technology Stack

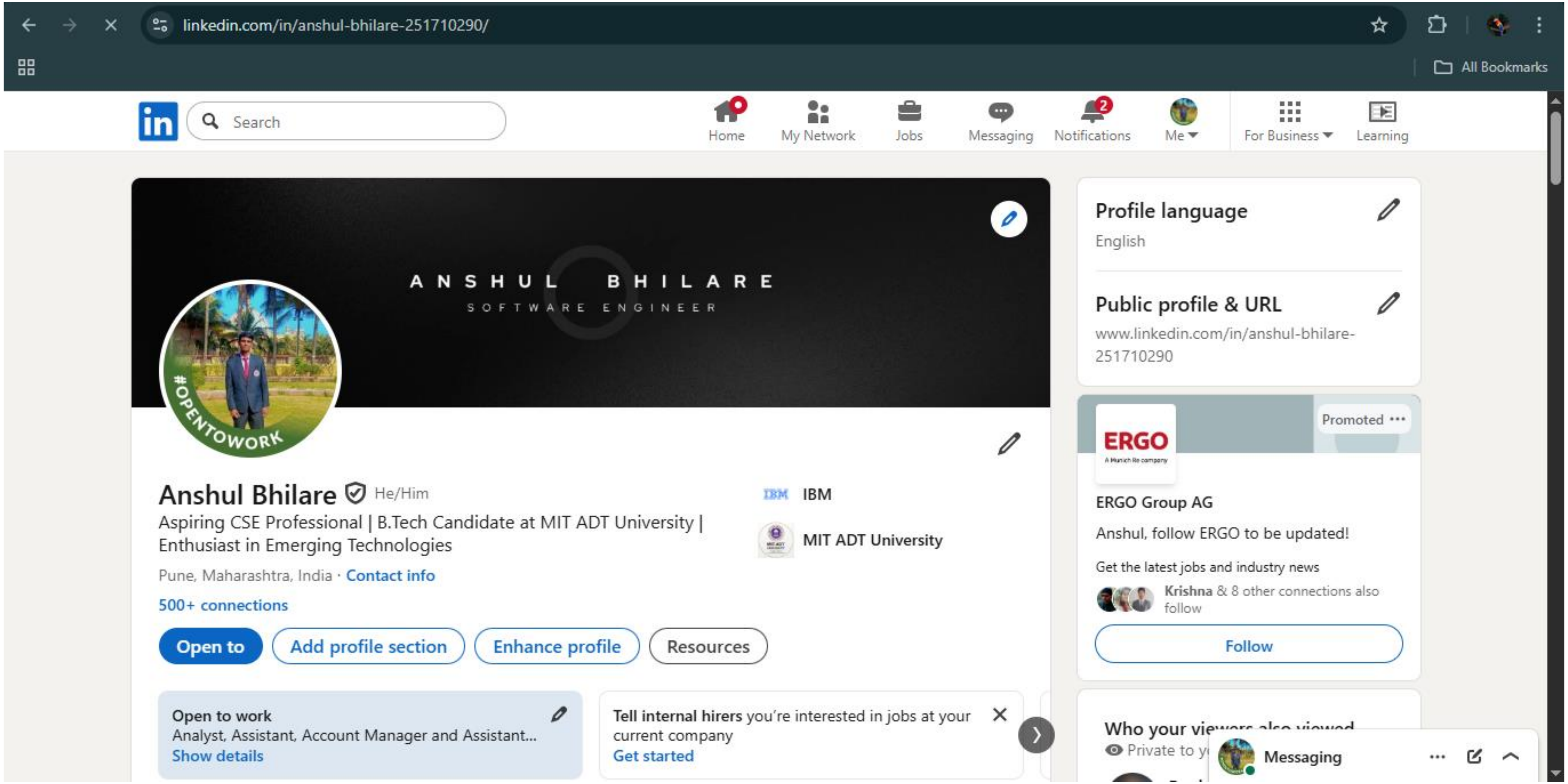
- **Frontend**
- **React.js** / **Next.js** (Web)
- **React Native** (optional mobile app)
- **TailwindCSS** for UI styling
- **Speech-to-Text API** (Web Speech API)
- **Backend**
- **Node.js** with **Express/Fastify**
- **REST APIs** / **GraphQL**
- **JWT Authentication**
- **Multer** for file uploads (reports)
- **AI / ML Models**
- **LLM APIs:** OpenAI / Gemini / LLaMA hosted model
- **OCR:** Tesseract OCR or Google Vision API
- **Symptom Classification:** lightweight ML model (XGBoost / Scikit-learn)
- **Rule Engine:** Custom red-flag safety rules (JSON + Node service)

Domain-Specific Deep Dive

- **Frontend**: React (PWA) with file upload, voice capture (Web Speech), chat UI, language toggle.
- **API / Orchestrator**: Node.js + Express/Fastify. Manages flow & invokes services.
- **OCR**: Tesseract (open) or Google Vision (production).
- **Parser**: Maps OCR tokens → test names, values, units, reference ranges.
- **Rule Engine**: Deterministic JSON rules for red flags (always overrides LLM).
- **Classifier**: Light ML model (XGBoost / sklearn) for predicting urgency from structured features.
- **LLM**: Prompt-engineered model for natural explanations + next steps (OpenAI/Gemini or local Llama variant).
- **Storage**: PostgreSQL for structured data, S3 for files, Redis for cache.
- **Admin**: Dashboard to view flagged cases and accept handoffs.

Prototype / UI Screens

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All Bookmarks
Search
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ANSHUL BHILARE
SOFTWARE ENGINEER
#OPENTOWORK
Anshul Bhilare
He/Him
Aspiring CSE Professional | B.Tech Candidate at MIT ADT University | Enthusiast in Emerging Technologies
Pune, Maharashtra, India · Contact info
500+ connections
IBM
MIT ADT University
Open to
Add profile section
Enhance profile
Resources
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Show details
Tell internal hirers you're interested in jobs at your current company
Get started
Profile language
English
Public profile & URL
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Promoted
ERGO
ERGO Group AG
Anshul, follow ERGO to be updated!
Get the latest jobs and industry news
Krishna & 8 other connections also follow
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Messaging

Innovation & Uniqueness

- **Multi-Domain Medical Intelligence**
- Doctor AI uniquely blends **symptom investigation**, **medical report understanding**, **medicine guidance**, and **risk prediction** into one unified LLM pipeline—unlike typical symptom checkers that give only basic suggestions.
- **2. Hybrid LLM + Medical-Knowledge Graph**
- Uses a **dual-engine approach**:
- LLM for natural conversation + reasoning
- Medical ontology (ICD-10, drug database) for **accurate disease mapping**
  This drastically reduces hallucinations.
- **3. Context-Aware Dynamic Triage**
- Provides **emergency severity scoring** based on symptoms, history, vitals, and report data—making it more actionable than simple chatbot responses.

Impact Analysis

- **1. Improved Healthcare Accessibility**
- Doctor AI offers instant, 24×7 medical assistance, especially for **rural users, students, senior citizens**, and those who lack quick access to doctors. This significantly reduces the dependency on physical consultations for minor issues.
- **2. Reduction in Self-Medication Risks**
- Instead of blind Googling or guessing medicines, the system gives **safe, evidence-based guidance**, lowering chances of wrong drug usage or harmful combinations.
- **3. Faster Pre-diagnosis & Early Detection**
- The AI can identify red-flag symptoms, abnormal report values, and high-risk patterns, leading to **earlier treatment opportunities** and preventing complications.
- **4. Lower Clinical Workload**
- Basic queries, report explanations, lifestyle suggestions, and triage scoring are automated. Doctors can focus on **critical cases**, reducing consultation time by **30–40%**.

Future Enhancements & Challenges Faced

1. Wearable Device Integration
Sync real-time health data (SpO_2, Heart Rate, BP, Sleep Patterns) for **continuous monitoring & instant alerts**.
2. Advanced Report Scanner (OCR + Vision AI)
Support X-rays, ECGs, MRIs using open-source medical imaging models for **automated anomaly detection**.
3. Personalized Health Records (PHR) Vault
User-specific encrypted health profiles, medication history, allergies, and prescriptions.
4. Multi-Lingual Voice Assistant
Support for **10+ Indian languages**, allowing rural users to interact fully via voice.
5. Doctor Collaboration Portal
Doctors can verify AI outputs and provide corrections—creating a **human-in-the-loop** learning cycle.
6. Integration with ABDM / Ayushman Bharat APIs
To fetch digital health records and offer connected public healthcare services.

Conclusion & Thank You

Deployed Link:
https://doctoraiviaaanshul.netlify.app/

LinkedIn: https://www.linkedin.com/in/anshul-bhilare-251710290/

Portfolio: https://anshulbhilareportfolio.netlify.app/

GitHub Link: https://github.com/anshbytecode/DoctorAiadvance

This Topic got selected at **IIT Delhi** , and we were **Finalist at IIT Delhi Hackathon**.
We have made this project frontend,backend and made it upto web3 and integrated into android studio as well, and made apk out of it and now it is working on mobile phones as well. Also participated in various competitions at **IIT Bombay**.